

DATABASE PROJECT REPORT

UNIVERSITY RESEARCH & LAB ASSET TRACKER

A. PROJECT IDENTITY

Item	Details
Project ID & Title	Project #1: University Research & Lab Asset Tracker
Team Name	G1
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Project Focus	Research grants, laboratory equipment, chemical inventory, personnel access control, EER specialization, and certification-based SQL Trigger integrity.

B. DOMAIN GLOSSARY & TAXONOMY

Domain Term	Database Semantic Definition	Representative Instance
Personnel	Common supertype representing people who may access university laboratories; specialized into Faculty or Student.	PER-00001
Research Lab	A university laboratory that stores equipment and chemicals and has an assigned hazard level.	LAB-001
Equipment	A physical research asset assigned to exactly one laboratory.	EQP-001
Chemical	A stored laboratory substance with quantity, location, and hazard classification.	CHM-001
Certification	A safety qualification defining an authorized hazard-access level.	CERT-High
Research Grant	A funded research project led by one Faculty PI and involving participating personnel.	GRANT-001
Access Log	An entry/exit audit record linking personnel to a lab and optionally an equipment item or chemical.	LOG-00001

1. SYSTEM SCOPE & PROBLEM STATEMENT

1.1 Problem Statement & Objectives

Universities operate multiple research laboratories serving faculty and students, using high-value equipment and chemicals while receiving funding from multiple research grants. When asset, chemical, grant, and access information is maintained in paper records or separate spreadsheets, traceability and safety control become difficult.

- It is difficult to control who is accessing hazardous chemicals or restricted laboratory areas.
- Equipment and chemicals may not be clearly associated with a specific laboratory or research activity, making audits difficult.
- There is no mandatory database-level rule ensuring that only personnel with appropriate and valid safety certifications can access hazardous materials.
- Personnel, grants, laboratories, and laboratory assets are insufficiently linked for centralized reporting.

The objective is to design a relational database named "University Research & Lab Asset Tracker" for centralized management of research grants, laboratory equipment, chemical inventory, and access logs. The central integrity challenge is to ensure that only personnel with valid, unexpired certifications may access materials or areas with the corresponding hazard level; this rule is intended to be enforced at the database layer through an SQL Trigger.

1.2 System Scope

Scope Area	Description
In Scope	Departments; Personnel (Faculty/Students); safety certifications; laboratories; equipment; chemicals; research grants; grant membership; access logs.
Out of Scope	Full university financial accounting; academic course registration; complete university HR management.

1.3 Functional Requirements

ID	Functional Requirement
FR1	Administrator registers new laboratory equipment and assigns it to a specific Lab.
FR2	Faculty members create and manage research grant records, including funding organization, grant amount, and project duration.
FR3	The system records each personnel laboratory access event with timestamp and Entry/Exit access type.
FR4	The database validates safety certification before access to hazardous laboratories or chemicals; enforcement is through an SQL Trigger.
FR5	The system tracks chemical inventory, stock quantity, storage location, and hazard classification.
FR6	Multiple Personnel members can participate in a Grant with roles such as PI, Co-PI, or Research Assistant.
FR7	The system supports audit/report queries such as unauthorized access attempts, equipment by lab, and certification-related information.

1.4 Non-Functional Requirements

ID	Requirement
NFR1 – Security	Certification and personnel identification data require strict access control; encryption at rest is recommended for sensitive fields.
NFR2 – Performance	Access-log audit queries should return in under 200 ms with up to 100,000 records.
NFR3 – Availability	The system should be available 24/7 because laboratories may be accessed outside normal working hours.
NFR4 – Data Integrity	Certification-based hazardous access control must be enforced at the database layer and not depend entirely on the application.

2. ENTERPRISE BUSINESS RULES & INTEGRITY CONSTRAINTS

2.1 Business Rules Subsystem Mapping & Traceability Matrix

Subsystem / Functional Area	Rule Range	Count	Associated Entities	Core Constraint
Personnel & Certification	BR1–BR5	5	PERSONNEL, FACULTY, STUDENT, DEPARTMENT, CERTIFICATION, PERSON CERTIFICATION	Disjoint/Partial specialization; certification validity
Laboratories, Equipment & Chemicals	BR6–BR8	3	LAB, EQUIPMENT, CHEMICAL, PERSONNEL, CERTIFICATION	1:N ownership/storage; hazard-level access rule
Research Grants	BR9–BR12	4	RESEARCH GRANT,	Exactly one Faculty

			FACULTY, PERSONNEL, GRANT_MEMBER	PI; N:N membership; amount/date checks
Access Log	BR13–BR15	3	ACCESS_LOG, PERSONNEL, LAB, EQUIPMENT, CHEMICAL, CERTIFICATION	FK integrity; Entry/Exit status; trigger-based authorization

2.2 Detailed Enterprise Business Rules Specification (BR1 – BR15)

Rule ID	Functional Category	Enterprise Database Business Rule & Integrity Constraint
BR1	Personnel Identity & Specialization	Each PERSONNEL has a unique person_id and may specialize as FACULTY or STUDENT. The specialization is disjoint and partial.
BR2	Faculty Department	Each FACULTY belongs to exactly one DEPARTMENT; a DEPARTMENT may have many FACULTY members.
BR3	Student Advisor	Each STUDENT may have at most one FACULTY advisor; a FACULTY member may advise many STUDENTS.
BR4	Personnel Certification	PERSONNEL and CERTIFICATION have an N:N relationship through PERSON_CERTIFICATION; each issuance records issue_date and expiry_date.
BR5	Certification Validity	A certification is valid only when the current date falls within [issue_date, expiry_date].
BR6	Lab Equipment	Each EQUIPMENT item belongs to exactly one LAB; a LAB may contain zero or more EQUIPMENT items.
BR7	Lab Chemical	Each CHEMICAL is stored in exactly one LAB; a LAB may store zero or more CHEMICALS.
BR8	Hazard Access	If a LAB/CHEMICAL hazard level is not 'None', only PERSONNEL with a valid certification at the corresponding or higher level may access it; enforced by a BEFORE INSERT trigger on ACCESS_LOG.
BR9	Grant PI	Each RESEARCH_GRANT has exactly one Principal Investigator, and the PI must be FACULTY.
BR10	Faculty as PI	A FACULTY member may be PI of multiple RESEARCH_GRANTS.
BR11	Grant Membership	A GRANT may have many participating PERSONNEL with roles through GRANT_MEMBER; a PERSONNEL member may participate in multiple GRANTS.
BR12	Grant Validity	Grant amount must be greater than 0 and end_date must be later than start_date.
BR13	Access Log References	Each ACCESS_LOG references an existing PERSONNEL and LAB; it may optionally reference a specific EQUIPMENT item and/or CHEMICAL.
BR14	Access Authorization	Each access event has access_type Entry or Exit; authorized_flag is determined

		from valid certification relative to the related Lab/Chemical hazard level.
BR15	Central DB Challenge	The system must not insert an ACCESS_LOG with authorized_flag = TRUE when the PERSONNEL lacks the corresponding valid certification; this will be implemented by SQL Trigger in Phase 3.

3. DATABASE DESIGN

3.1 Conceptual Model: Enhanced Entity-Relationship (EER) Diagram

The conceptual model uses EER specialization/generalization for PERSONNEL → FACULTY / STUDENT, associative entities for certification and grant membership, and relationships connecting laboratories, assets, grants, and access events. The original Phase 1 EER diagram is retained below.

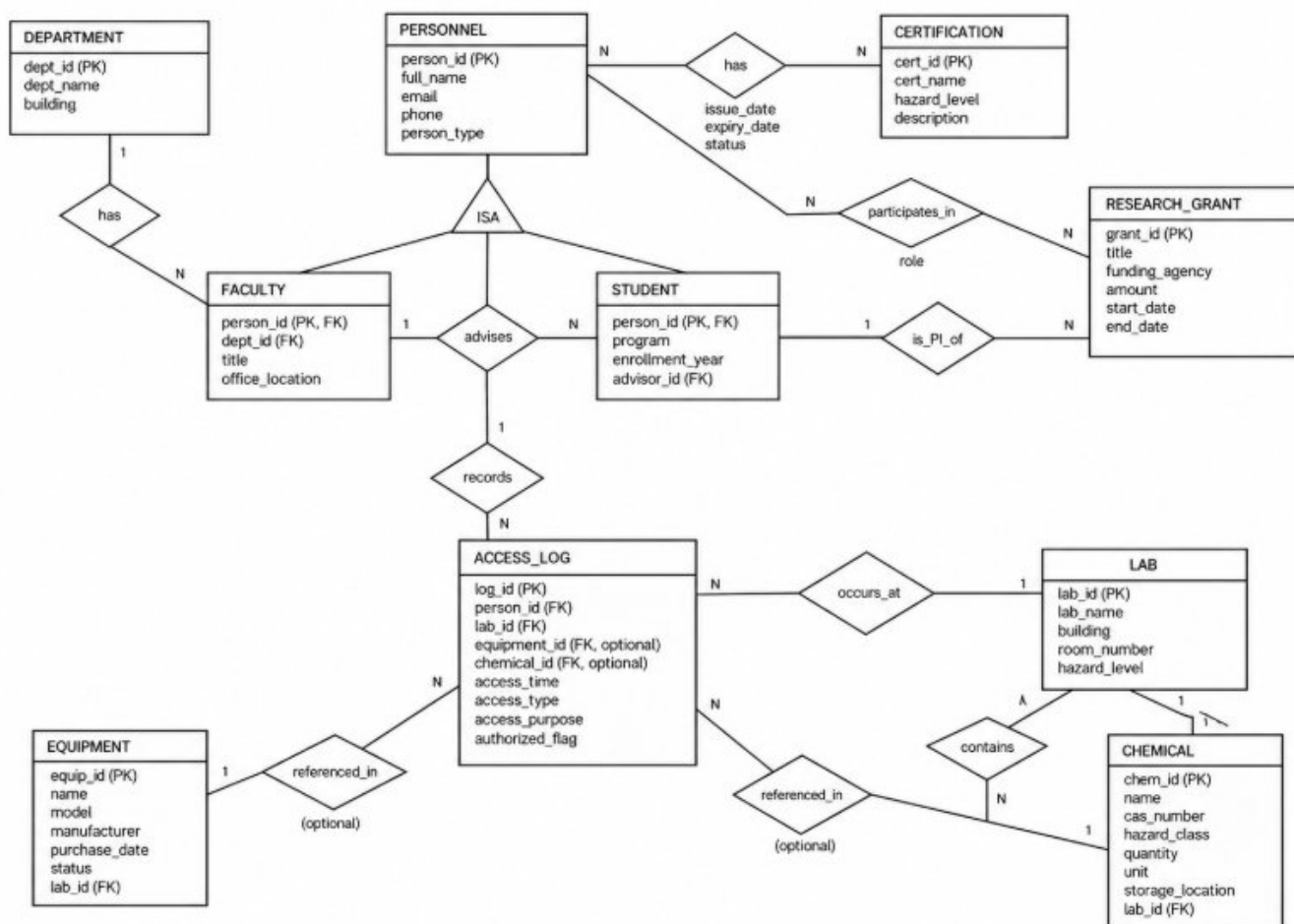


Figure 1: Comprehensive EER Conceptual Schema

3.2 Entity Overview

Entity	Primary Key (PK)	Purpose
DEPARTMENT	dept_id	Department/unit responsible for managing faculty members.
PERSONNEL	person_id	Common supertype for all personnel.
FACULTY	person_id (PK, FK)	Faculty subtype; may serve as grant PI

		and advise students.
STUDENT	person_id (PK, FK)	Student subtype; may have a Faculty advisor.
CERTIFICATION	cert_id	Safety certification type and authorized hazard-access level.
PERSON_CERTIFICATION	person_id, cert_id	Associative entity recording certification issuance and validity.
LAB	lab_id	Laboratory with its own hazard level.
EQUIPMENT	equipment_id	Equipment assigned to a specific Lab.
CHEMICAL	chemical_id	Chemical stored in a Lab with hazard classification.
RESEARCH GRANT	grant_id	Research grant with one Faculty PI.
GRANT_MEMBER	grant_id, person_id	Associative entity between Grant and participating Personnel, including role.
ACCESS_LOG	log_id	Lab entry/exit audit record with optional Equipment/Chemical reference and authorization flag.

3.3 Relationship Semantics & Referential Constraints

#	Entity Relationship	Cardinality & Participation	Business Semantics	BR
R1	DEPARTMENT → FACULTY	1 : 0..N	Each Faculty belongs to exactly one Department; a Department may manage many Faculty.	BR2
R2	FACULTY → STUDENT (advises)	1 : 0..N / Student 0..1	A Student may have at most one Faculty advisor; a Faculty may advise many Students.	BR3
R3	PERSONNEL ↔ CERTIFICATION	N : N via PERSON_CERTIFICATION	Certification issuance stores issue_date and expiry_date.	BR4–BR5
R4	LAB → EQUIPMENT	1 : 0..N	Each Equipment item belongs to exactly one Lab.	BR6
R5	LAB → CHEMICAL	1 : 0..N	Each Chemical is stored in exactly one Lab.	BR7
R6	FACULTY → RESEARCH_GRANT (PI)	1 : 0..N; Grant exactly 1 PI	Each Grant has exactly one Faculty PI; Faculty may lead multiple Grants.	BR9–BR10
R7	RESEARCH_GRANT ↔ PERSONNEL	N : N via GRANT_MEMBER	Personnel may participate in multiple Grants with assigned roles.	BR11
R8	PERSONNEL → ACCESS_LOG	1 : 0..N	Every access log references one valid Personnel member.	BR13
R9	LAB → ACCESS_LOG	1 : 0..N	Every access log references one Lab.	BR13
R10	EQUIPMENT → ACCESS_LOG	1 : 0..N (optional from log)	An access event may optionally reference one Equipment item.	BR13
R11	CHEMICAL → ACCESS_LOG	1 : 0..N (optional from log)	An access event may optionally reference one Chemical.	BR13

R12	PERSONNEL → FACULTY / STUDENT	Disjoint, Partial	A Personnel member may be Faculty or Student, but not both; either subtype may be absent.	BR1
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3.4 Design Assumptions

- A Personnel member cannot be both Faculty and Student at the same time (disjoint specialization).
- A Grant has only one PI at a time; PI change history is outside Phase 1.
- Each Access Log is associated with at most one Equipment item OR one Chemical; neither is mandatory.
- Lab, Chemical, and Certification use a consistent hazard scale (for example: None, Low, Medium, High) so the Phase 3 trigger can compare levels directly.

4. PHASE 2 – LOGICAL DESIGN & NORMALIZATION

Phase 2 converts the conceptual EER design into a relational schema, defines foreign-key referential actions, and verifies the relations through 1NF, 2NF, 3NF, and BCNF.

4.1 Logical Schema Mapping

Mapping Rules Applied:

Rule 1 — Strong Entity: Each strong entity maps to a separate relation table. The entity's key becomes the table's PK.

Rule 2 — Single-valued Attributes: Each single-valued attribute becomes a column in the table.

Rule 3 — Binary 1:N Relationship: The PK of the '1' side is added to the 'N' side table as a FK. Example: EQUIPMENT.lab_id → LAB.lab_id.

Rule 4 — Binary N:N Relationship: A new association table is created with FKs from both sides. The composite PK consists of those FKs. Applied to PERSON_CERTIFICATION and GRANT_MEMBER.

Rule 5 — EER Specialization (Method 8A): For DISJOINT PARTIAL specialization (PERSONNEL → FACULTY/STUDENT): retain the supertype table PERSONNEL and create separate subtype tables FACULTY and STUDENT. Each subtype's PK is a FK referencing PERSONNEL with ON DELETE CASCADE. Chosen because: (a) avoids excessive NULL values, (b) supports partial specialization (personnel in neither subtype remain valid), (c) enables efficient subtype-specific queries.

Rule 6 — Optional FK (Nullable): Optional relationships are represented by nullable FKs. Example: ACCESS_LOG.equipment_id and ACCESS_LOG.chemical_id may be NULL when the event is not linked to a specific item.

Full Relational Schema (Text Notation):

DEPARTMENT (dept_id [PK], dept_name, building)

PERSONNEL (person_id [PK], full_name, email, phone, person_type)

FACULTY (person_id [PK, FK*→PERSONNEL], dept_id [FK*→DEPARTMENT, null], tenure_status, office_location)

STUDENT (person_id [PK, FK*→PERSONNEL], program, enrollment_year, advisor_id [FK*→FACULTY, null])

CERTIFICATION (cert_id [PK], cert_name, hazard_level, description)

PERSON_CERTIFICATION (person_id [PK, FK*→PERSONNEL], cert_id [PK, FK*→CERTIFICATION], issue_date, expiry_date, status)

LAB (lab_id [PK], lab_name, building, room_number, hazard_level)

EQUIPMENT (equipment_id [PK], name, model, serial_number, purchase_date, status, lab_id [FK*→LAB])

CHEMICAL (chemical_id [PK], name, cas_number, hazard_class, quantity, unit, storage_location, lab_id [FK*→LAB])

RESEARCH_GRANT (grant_id [PK], title, funding_agency, amount, start_date, end_date, pi_id [FK*→FACULTY])

GRANT_MEMBER (grant_id [PK, FK*→RESEARCH_GRANT], person_id [PK, FK*→PERSONNEL], role_on_grant)

ACCESS_LOG (log_id [PK, AUTO_INCREMENT], person_id [FK*→PERSONNEL], lab_id [FK*→LAB], equipment_id [FK*→EQUIPMENT, null], chemical_id [FK*→CHEMICAL, null], access_timestamp, access_type, authorized_flag)

Foreign Key Summary:

Child Table	FK Column	References (Parent)	ON DELETE	Business Rule
FACULTY	dept_id	DEPARTMENT(dept_id)	SET NULL	BR2
FACULTY	person_id	PERSONNEL(person_id)	CASCADE	BR1
STUDENT	person_id	PERSONNEL(person_id)	CASCADE	BR1
STUDENT	advisor_id	FACULTY(person_id)	SET NULL	BR3
PERSON_CERTIFICATION	person_id	PERSONNEL(person_id)	CASCADE	BR4
PERSON_CERTIFICATION	cert_id	CERTIFICATION(cert_id)	RESTRICT	BR4
EQUIPMENT	lab_id	LAB(lab_id)	RESTRICT	BR6
CHEMICAL	lab_id	LAB(lab_id)	RESTRICT	BR7
RESEARCH_GRANT	pi_id	FACULTY(person_id)	RESTRICT	BR9
GRANT_MEMBER	grant_id	RESEARCH_GRANT(grant_id)	CASCADE	BR11
GRANT_MEMBER	person_id	PERSONNEL(person_id)	CASCADE	BR11
ACCESS_LOG	person_id	PERSONNEL(person_id)	RESTRICT	BR13
ACCESS_LOG	lab_id	LAB(lab_id)	RESTRICT	BR13
ACCESS_LOG	equipment_id	EQUIPMENT(equipment_id)	SET NULL	BR13
ACCESS_LOG	chemical_id	CHEMICAL(chemical_id)	SET NULL	BR13

4.2 Normalization Verification

Normalization eliminates data redundancy and ensures integrity. We verify each relation from 1NF through BCNF using Functional Dependency (FD) analysis.

1. **1NF:** All attributes are atomic (no repeating groups, no multi-valued attributes).
2. **2NF:** Achieves 1NF AND no partial dependency: every non-key attribute depends on the WHOLE PK (only relevant for composite PKs).
3. **3NF:** Achieves 2NF AND no transitive dependency: for every FD $X \rightarrow Y$, X must be a superkey or Y must be a prime attribute.
4. **BCNF:** Stricter than 3NF: for every FD $X \rightarrow Y$, X must be a superkey. Eliminates all redundancy based on FDs.

Table: DEPARTMENT

Primary Key / Candidate Keys	PK: dept_id Candidate Keys: dept_id
Functional Dependencies (FDs)	dept_id -> dept_name, building

Normalization Proof	1NF: All attributes atomic. 2NF: Single-attr PK, no partial deps. 3NF: No transitive deps. BCNF: Only determinant is dept_id (superkey). Conclusion: Achieves BCNF.
Conclusion	DEPARTMENT achieves BCNF.

Table: PERSONNEL

Primary Key / Candidate Keys	PK: person_id Candidate Keys: person_id, email
Functional Dependencies (FDs)	person_id -> full_name, email, phone, person_type email -> person_id, full_name, phone, person_type
Normalization Proof	1NF: All attributes atomic. 2NF: Single-attr PK. 3NF: No transitive deps. BCNF: Both determinants (person_id, email) are candidate keys = superkeys. Conclusion: Achieves BCNF.
Conclusion	PERSONNEL achieves BCNF.

Table: FACULTY

Primary Key / Candidate Keys	PK: person_id Candidate Keys: person_id
Functional Dependencies (FDs)	person_id -> dept_id, tenure_status, office_location
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: dept_id is FK, does not cause transitive dependency in FACULTY table as department name is not stored here. Conclusion: Achieves BCNF.
Conclusion	FACULTY achieves BCNF.

Table: STUDENT

Primary Key / Candidate Keys	PK: person_id Candidate Keys: person_id
Functional Dependencies (FDs)	person_id -> program, enrollment_year, advisor_id
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: advisor_id is FK, does not cause transitive dependency. Conclusion: Achieves BCNF.
Conclusion	STUDENT achieves BCNF.

Table: CERTIFICATION

Primary Key / Candidate Keys	PK: cert_id Candidate Keys: cert_id, cert_name
Functional Dependencies (FDs)	cert_id -> cert_name, hazard_level, description cert_name -> cert_id, hazard_level, description
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: Both determinants (cert_id, cert_name) are candidate keys. Conclusion: Achieves BCNF.
Conclusion	CERTIFICATION achieves BCNF.

Table: PERSON_CERTIFICATION

Primary Key / Candidate Keys	PK: (person_id, cert_id) Candidate Keys: (person_id, cert_id)
Functional Dependencies (FDs)	(person_id, cert_id) -> issue_date, expiry_date, status
Normalization Proof	1NF: Atomic. 2NF: Single person_id or cert_id does not determine issue_date / expiry_date - no partial dependency. 3NF & BCNF: Only one FD, left side is composite PK (superkey). Conclusion: Achieves BCNF.
Conclusion	PERSON_CERTIFICATION achieves BCNF.

Table: LAB

Primary Key / Candidate Keys	PK: lab_id Candidate Keys: lab_id, lab_name, (building, room_number)
Functional Dependencies (FDs)	lab_id -> lab_name, building, room_number, hazard_level (building, room_number) -> lab_id, lab_name, hazard_level

Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: Every determinant is a candidate key (superkey). Conclusion: Achieves BCNF.
Conclusion	LAB achieves BCNF.

Table: EQUIPMENT

Primary Key / Candidate Keys	PK: equipment_id Candidate Keys: equipment_id, serial_number (khi NOT NULL)
Functional Dependencies (FDs)	equipment_id -> name, model, serial_number, purchase_date, status, lab_id
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: No transitive dependency. lab_id is FK but lab name/details are not stored here. Conclusion: Achieves BCNF.
Conclusion	EQUIPMENT achieves BCNF.

Table: CHEMICAL

Primary Key / Candidate Keys	PK: chemical_id Candidate Keys: chemical_id, cas_number (khi NOT NULL)
Functional Dependencies (FDs)	chemical_id -> name, cas_number, hazard_class, quantity, unit, storage_location, lab_id
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: No transitive dependency. lab_id is FK. Conclusion: Achieves BCNF.
Conclusion	CHEMICAL achieves BCNF.

Table: RESEARCH_GRANT

Primary Key / Candidate Keys	PK: grant_id Candidate Keys: grant_id
Functional Dependencies (FDs)	grant_id -> title, funding_agency, amount, start_date, end_date, pi_id
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK. 3NF & BCNF: pi_id is FK, PI name is not stored here. No transitive dependency. Conclusion: Achieves BCNF.
Conclusion	RESEARCH GRANT achieves BCNF.

Table: GRANT_MEMBER

Primary Key / Candidate Keys	PK: (grant_id, person_id) Candidate Keys: (grant_id, person_id)
Functional Dependencies (FDs)	(grant_id, person_id) -> role_on_grant
Normalization Proof	1NF: Atomic. 2NF: Single grant_id or single person_id does not determine role_on_grant - no partial dependency. 3NF & BCNF: Only one FD, left side is composite PK. Conclusion: Achieves BCNF.
Conclusion	GRANT_MEMBER achieves BCNF.

Table: ACCESS_LOG

Primary Key / Candidate Keys	PK: log_id Candidate Keys: log_id
Functional Dependencies (FDs)	log_id -> person_id, lab_id, equipment_id, chemical_id, access_timestamp, access_type, authorized_flag
Normalization Proof	1NF: Atomic. 2NF: Single-attr PK (AUTO_INCREMENT). 3NF & BCNF: authorized_flag is determined by trigger based on business logic, not an FD in schema. No transitive dependency. Conclusion: Achieves BCNF.
Conclusion	ACCESS_LOG achieves BCNF.

Normalization Summary:

Table	1NF	2NF	3NF	BCNF	PK Type
DEPARTMENT	Pass	Pass	Pass	Pass	Single
PERSONNEL	Pass	Pass	Pass	Pass	Single

FACULTY	Pass	Pass	Pass	Pass	Single
STUDENT	Pass	Pass	Pass	Pass	Single
CERTIFICATION	Pass	Pass	Pass	Pass	Single
PERSON_CERTIFICATION	Pass	Pass	Pass	Pass	Composite
LAB	Pass	Pass	Pass	Pass	Single
EQUIPMENT	Pass	Pass	Pass	Pass	Single
CHEMICAL	Pass	Pass	Pass	Pass	Single
RESEARCH_GRANT	Pass	Pass	Pass	Pass	Single
GRANT_MEMBER	Pass	Pass	Pass	Pass	Composite
ACCESS_LOG	Pass	Pass	Pass	Pass	Single (AUTO_INCREMENT)